

Base on contract.md I provide, please implement the Express.js route file (routes/products.js) that meets the specified requirements:

1. Accepts a query parameter for 'category' (filter by it when given, return all products when omitted).
2. Reads products from ./data/products.json using fs (not a database).
3. Filters the array in JavaScript by category if the param is provided.
4. Returns the exact JSON shape from the contract above.
5. Catches file read errors and returns a 500 response.
6. Add comments that map each block to the sequence diagram steps - .